(1) Knowledge Inversion Phase

🎯 Task Instruction and Inputs
Inferred user-level goals and input specifications from the extracted information.

Match strategic patterns to possible task-level objectives.

🧠 Strategy Description
Further summarize the high-level plan that outlines the intent of the workflow.

Distill strategic insights and summarize high-level decision flow.

🧠 Strategy Reasoning Extraction
Infer reasoning traces that connect pseudo-code to strategic intent.

*Interpret structural patterns to recover the **underlying reasoning logic**.*

🧩 Skeleton Pseudo-code
Simplified structural code capturing workflow topology.

Abstract structural relations by removing parameter bindings and low-level details.

🧩 Full Pseudo-code
Reconstructed code containing modules and parameterized components.

Rule-based structural regularization

⚙️ Raw ComfyUI Workflow Collection
Original executable graphs collected from real-world workflows.

Used to train

Used to train

Symmetry Operations

(2) Inference Phase

🎯 Task Instruction and Inputs

I want to create an image from text prompt, apply my chosen style/character presets to control the look, and then upscale it to a crisp, high-resolution final result suitable for large displays or print.

High-level Strategy Planner

Self-Refinement

🧠 **Strategy:**
Leverage a LoRA-augmented diffusion backbone to generate an image from text ..... **encode positive and negative prompts using CLIP**, **stack multiple LoRAs** onto both the UNet and CLIP to align style and semantics ..... sample a base image in latent space, decode with a VAE, and finally enforce structure and detail **via a tile-aware, diffusion-assisted upscaling stage** .....

Pseudo-code Generator

Self-Refinement

🧩 **Pseudo-code**

```
MODEL_11 = UNETLoader()
CLIP_13 = DualCLIPLoader()
MODEL_12, CLIP_12 = LoraLoader(model=MODEL_11,
clip=CLIP_13)
MODEL_8, CLIP_8 = LoraLoader(model=MODEL_12,
clip=CLIP_12)
......
```

Parameter Filler → **Rule-based Reconstruction**



⚙️ **Complete ComfyUI Workflow**